

Untargeted Metabolomics - Materials and Methods

1. Metabolites Extraction

Tissue sample

Tissues (20 mg) were individually grounded with liquid nitrogen and the homogenate was resuspended with prechilled 90% methanol by well vortex. The samples were incubated on ice for 5 min and then were centrifuged at 15,000 rpm, 4°C for 20 min. Some of supernatant was diluted to final concentration containing 53% methanol by LC-MS grade water. The samples were subsequently transferred to a fresh Eppendorf tube and then were centrifuged at 15000 rpm, 4°C for 20 min. Finally, the supernatant was injected into the LC-MS/MS system analysis^[1].

Liquid sample

The samples (100 µL) were placed in the EP tubes and resuspended with prechilled methanol by well vortex. Then the samples were incubated on ice for 5 min and centrifuged at 15,000 rpm, 4°C for 20 min. Some of supernatant was diluted to final concentration containing 53% methanol by LC-MS grade water. The samples were subsequently transferred to a fresh Eppendorf tube and then were centrifuged at 15000 rpm, 4°C for 20 min. Finally, the supernatant was injected into the LC-MS/MS system analysis^[2-3].

Cell or bacteria sample

The samples were placed in the EP tubes and resuspended with prechilled 80% methanol by well vortex. Then the samples were melted on ice and whirled for 30 s. After water bath sonication for 15 min, they were centrifuged at 15,000 rpm, 4°C for 20 min. The supernatant was freeze-dried and dissolved with 10% methanol. Finally, the solution was injected into the LC-MS/MS system analysis^[4-5].

Cell or bacteria culture medium sample

The samples (1 mL) were freeze-dried and resuspended with prechilled 80% methanol by well vortex. Then the samples were incubated on ice for 5 min and centrifuged at 15,000 rpm, 4°C for 20 min. Some of supernatant was diluted to final concentration containing 53% methanol by LC-MS grade water. The samples were subsequently

transferred to a fresh Eppendorf tube and then were centrifuged at 15000 rpm, 4°C for 20 min. Finally, the supernatant was injected into the LC-MS/MS system analysis.

Soil sample

The samples (1g) were placed in the EP tubes and resuspended with prechilled 80% methanol by well vortex; Then the samples were incubated on ice for 5 min and centrifuged at 15,000 rpm, 4°C for 20 min, collected the supernatant, centrifuged at 15,000 rpm and 4°C for 20 minutes repeatedly; Finally, the solution was injected into the LC-MS/MS system analysis.

2. UHPLC-MS/MS Analysis (Key Point: Method ② only for HQ Untargeted Metabolomics Pro)

① UHPLC-MS/MS analyses were performed using a Vanquish UHPLC system (ThermoFisher, Germany) coupled with an Orbitrap Exploris™ 120 mass spectrometer (Thermo Fisher, Germany) in Novogene Co., Ltd. (Beijing, China). Samples were injected onto a Hypersil Goldcolumn (100×2.1 mm, 1.9μm) using a 12-min linear gradient at a flow rate of 0.2 mL/min. The eluents for the positive and negative polarity modes were eluent A (0.1% FA in Water) and eluent B (Methanol). The solvent gradient was set as follows: 2% B, 1.5 min; 2-85% B, 3 min; 85-100% B, 10 min; 100-2% B, 10.1 min; 2% B, 12 min. Orbitrap Exploris™ 120 mass spectrometer was operated in positive/negative polarity mode with spray voltage of 3.5 kV, capillary temperature of 320°C, sheath gas flow rate of 35 psi and aux gas flow rate of 10 L/min, S-lens RF level of 60, Aux gas heater temperature of 350°C.

② UHPLC-MS/MS analyses were performed using a Vanquish UHPLC system (ThermoFisher, Germany) coupled with an Orbitrap Exploris™ 120 mass spectrometer (ThermoFisher, Germany) in Novogene Co., Ltd. (Beijing, China). Samples were injected onto an ACQUITY UPLC BEH Amide Column (100×2.1 mm, 1.7 μm) using a 12-min linear gradient at a flow rate of 0.2 mL/min. The eluents for the positive and negative polarity mode were eluent A (5 mM ammonium acetate in 90% ACN,) and eluent B (5 mM ammonium acetate in 50% ACN,). The solvent gradient was set as follows: 2% B, 1.5 min; 2-100% B, 7 min; 100% B, 9 min; 100-2%

B, 9.1 min; 2% B, 12 min. Orbitrap Exploris™ 120 mass spectrometer was operated in positive/negative polarity mode with spray voltage of 3.5 kV, capillary temperature of 320°C, sheath gas flow rate of 35 arb and aux gas flow rate of 10 arb, S-lens RF level of 60, Aux gas heater temperature of 350°C.

3. Data processing and metabolite identification

The data files generated by UHPLC-MS/MS were processed by XCMS to perform peak alignment, peak picking, and quantitation for each metabolite. Then, based on adduct ions and setting mass deviation to 10 ppm^[6], a comparison was made between these data and the self-built high-quality secondary spectrum database (NovoMetDB) to obtain results for metabolite identification. Metabolites with more than 50% missing values in the samples were first filtered out and the remaining missing values were imputed using KNN^[7]. After eliminating background ions according to blank samples, the raw quantitative results were further processed to obtain relative peak areas, which included normalization and filtering out compounds whose relative peak areas exhibited a coefficient of variation (CV) greater than 30% in the quality control (QC) samples^[8]. Relative peak areas = Raw quantitative value of samples/ (The sum of quantitative value of samples/ The sum of quantitative value of QC1). Data processing is based on the Linux operating system (CentOS version 6.6), using R and Python. For details on specific packages and software versions used, please refer to the README file included in the results.

4. Data Analysis

These metabolites were annotated using the KEGG database (<https://www.genome.jp/kegg/pathway.html>) , HMDB database (<https://hmdb.ca/metabolites>) and LIPIDMaps database (<http://www.lipidmaps.org/>). Principal components analysis (PCA) and Partial least squares discriminant analysis (PLS-DA) were performed at metaX^[9] (a flexible and comprehensive software for processing metabolomics data). We applied univariate analysis (t-test) to calculate the statistical significance (P-value). FC refers to the fold change, which is the ratio of the mean of

all biological repeated quantitative values of each metabolite in the comparison group. The differential metabolites were screened using VIP, P-value and FC. The specific standard outlined in Table 5.3 of the report. Volcano plots were used to filter metabolites of interest which based on $\log_2(\text{FoldChange})$ and $-\log_{10}(\text{p-value})$ of metabolites by ggplot2 in R language.

For clustering heat maps, the data were normalized using z-scores of the intensity areas of differential metabolites and were plotted by Pheatmap package in R language. The correlation between differential metabolites were analyzed by `cor()` in R language (method=pearson). Statistically significant of correlation between differential metabolites were calculated by `cor.mtest()` in R language. P-value < 0.05 was considered as statistically significant and correlation plots were plotted by corrplot package in R language. The functions of these metabolites and metabolic pathways were studied using the KEGG database. The metabolic pathways enrichment of differential metabolites was performed, when ratio were satisfied by $x/n > y/N$, metabolic pathway were considered as enrichment, when P-value of metabolic pathway < 0.05, metabolic pathway were considered as statistically significant enrichment.

Reference

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